

QUALITY METRIC ANALYSIS REPORT

Telecom Customer Quality & Risk Assessment

7,043 Total Records	26.54% Failure Rate	\$139,131 Monthly Impact	30.50% Revenue Impact Rate
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PostgreSQL · Power BI

1. EXECUTIVE SUMMARY

This report presents a comprehensive quality metric analysis of a telecom customer dataset comprising 7,043 records across 6 integrated data sources. The objective is to identify the root causes of service quality failure (customer churn), quantify financial impact, and prescribe data-driven corrective actions for the Quality Audit & Metric team.

1.1 Scope & Objective

- Dataset: 7,043 customer records · 6 integrated tables
- Objective: Monitor quality KPIs, diagnose failure root causes, predict risk trajectory, prescribe corrective actions
- Tools: PostgreSQL (data pipeline) · Power BI (dashboard)
- Analysis Levels: Descriptive → Diagnostic → Predictive → Prescriptive

1.2 Key Findings

Finding	Metric	Business Impact
Revenue quality gap	Revenue Failure Rate 30.50% vs Customer Failure Rate 26.54%	Churned customers pay \$13.17/month more than retained ones
High-risk demographic	Senior 65+ churn at 42.25% · Single segment at 32.96%	Highest-value segments at disproportionate risk
Service ecosystem gap	Fiber Optic + No Addons: 58.33% churn rate	Core service sold without protective layer — driving 46.38% competitor loss

1.3 Recommendation Overview

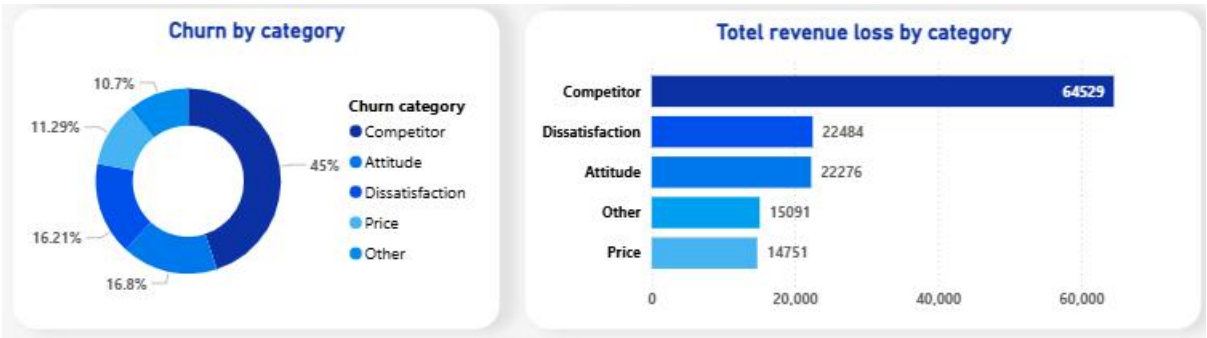
- REC 1 (Critical): Bundle Online Security + Tech Support for all Fiber Optic customers → projected churn reduction: **58.33% → 32.38%**
- REC 2 (High): Automated contract upgrade campaign at Month 2 → targeting Month-to-Month customers (**57.06%** churn in 0–6 months)
- REC 3 (Medium): Replace **Offer E** (52.92% churn) with Offer A framework (6.73% churn, 70 months avg tenure)

2. DESCRIPTIVE ANALYSIS — What Happened?

2.1 KPI Dashboard Summary

Total Customers	Quality Failure Rate	Monthly Revenue Impact	Revenue Failure Rate
7,043	26.54%	\$139,131	30.50%

2.2 Failure Distribution by Root Cause Category



1.1 Churn by category

The quality failure rate of 26.54% translates to a revenue impact of \$139,131/month. Notably, Revenue Failure Rate (30.50%) exceeds Customer Failure Rate (26.54%) by 3.96 percentage points — indicating that customers who leave are disproportionately high-value accounts.

Category	Customers Lost	Revenue Lost (\$)	% of Total Loss
Competitor	841 (45.0%)	\$64,529	46.38%
Attitude	314 (16.8%)	\$22,276	16.01%
Dissatisfaction	303 (16.2%)	\$22,484	16.16%
Price	211 (11.3%)	\$14,751	10.60%
Other	200 (10.7%)	\$15,091	10.85%

3. DIAGNOSTIC ANALYSIS — Why Did It Happen?

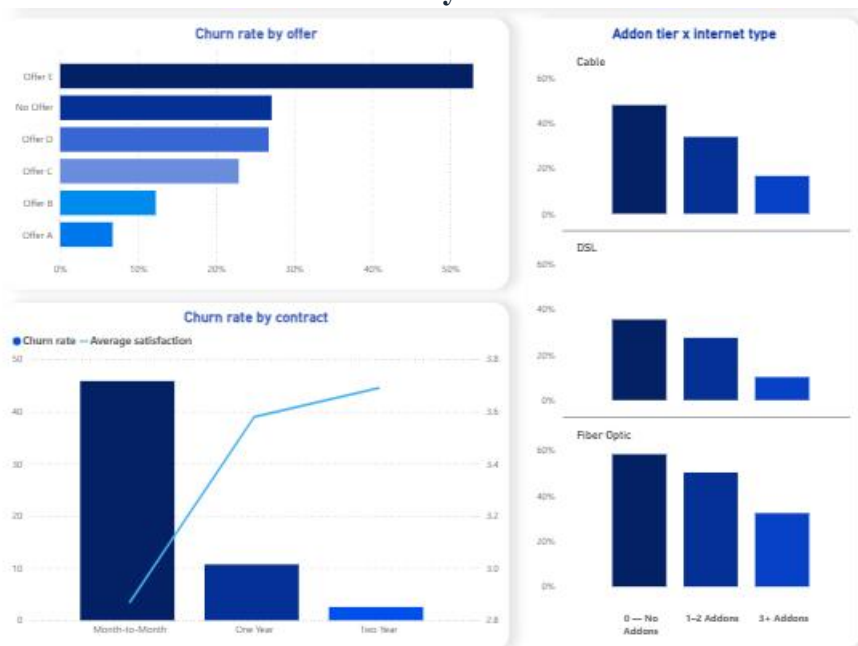
3.1 Customer Demographic Risk Segmentation



1.2 Demographic risk segmentation

- **Senior 65+** customers: 42.25% (Female) and 41.11% (Male) — highest churn rate across all age groups, despite paying \$80+/month
- **Single segment:** 32.96% churn rate · avg monthly charge \$71.45 — largest volume group with no natural lock-in factor
- Family with 1–3 dependents: only **3–5% churn** — confirming that family structure acts as the strongest natural retention mechanism

3.2 Service & Contract Root Cause Analysis



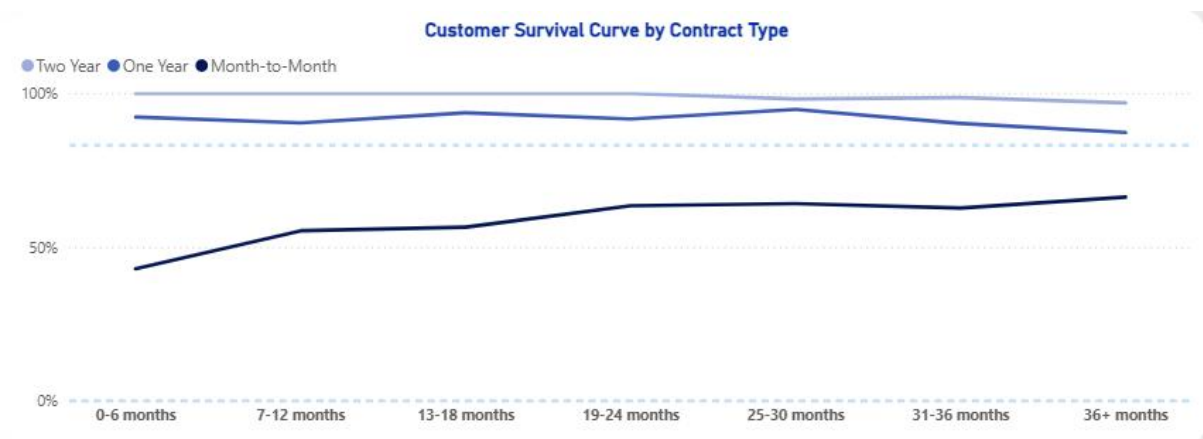
1.3 Service & Contract Root Cause Analysis

- Fiber Optic with no add-ons: **58.33% churn** — nearly 2x the overall rate. The core service is being sold without a protective ecosystem
- Online Security is the single most effective retention lever: reduces churn by **16.72 pts** (31.33% → 14.61%)
- Tech Support: reduces churn by **16.02 pts** (31.19% → 15.17%) — comparable impact to Online Security
- Streaming TV anomaly: **increases churn** from 24.33% to 30.07% — does not create ecosystem dependency
- **Offer E**: 52.92% churn within 3.7 months average tenure — a net-negative retention vehicle attracting price-sensitive customers
- Month-to-Month vs Two Year contract: **45.84% vs 2.55%** — contract type is the single most decisive retention factor

Risk Factor	Current Rate	Benchmark
Fiber Optic · No Add-ons	58.33% churn	32.38% (with 3+ add-ons)
Month-to-Month · 0–6 months	57.06% churn	0.00% (Two Year, same period)
Offer E subscribers	52.92% churn · 3.7 months	6.73% (Offer A · 70 months)

4. PREDICTIVE ANALYSIS — What Is Likely to Happen?

4.1 Customer Survival Curve by Contract Type



1.4 Service & Contract Root Cause Analysis

The survival curve reveals a stark divergence in retention trajectory by contract type. Two Year contract customers maintain near-100% retention through the first 6 months, while Month-to-Month customers drop to 42.94% retained in the same period. Critically, Month-to-Month customers never recover below 33% churn even at 36+ months — demonstrating that tenure length alone cannot compensate for contract type.

Tenure Stage	Month-to-Month	One Year	Two Year
0–6 months	42.94% retained	92.31% retained	100.00% retained
7–12 months	55.31% retained	90.43% retained	100.00% retained
13–24 months	56.49–63.48%	91.67–93.68%	100.00% retained
36+ months	66.26% retained	87.30% retained	96.94% retained

4.2 Projected Revenue Loss — No Intervention Scenario



1.5 Projected Revenue Loss — No Intervention Scenario

Based on current Fiber Optic churn rate of 58.33% with no corrective action, the projected trajectory indicates continued quarterly revenue deterioration. The following forward-looking risk estimates are derived from current churn rates applied to the active customer base:

Risk Segment	Monthly Revenue at Risk	Quarterly Projection
Fiber Optic · No Add-ons (active)	\$3,981/month total	Est. 12,00/quarter if unaddressed
Month-to-Month · 0–6 months (active)	\$1,662/month total M2M base	57.06% dropout rate projected to continue
Offer E subscribers (active)	\$993/month	52.92% projected to churn within 3.7 months

5. PRESCRIPTIVE ANALYSIS — What Should We Do?

Customer ID	Churn score	CLTV	Contract	Internet type	Addon count	Monthly charge	Retention priority
0052-YYNOT	72	5462	One Year	No internet services	0	20.55	Critical
0078-XZMHT	76	6322	Two Year	DSL	6	85.15	Critical
0141-YEAYS	77	5990	Two Year	Fiber Optic	2	86.45	Critical
0178-CIIKR	74	5949	Month-to-Month	No internet services	0	19.95	Critical
0206-OYVOC	77	5596	Two Year	No internet services	0	19.25	Critical
0230-WEQUW	77	5500	Two Year	DSL	5	56.60	Critical
0244-LGNFY	75	6342	Two Year	DSL	7	92.65	Critical
0248-PGHBZ	80	5857	Two Year	Fiber Optic	4	92.45	Critical
0257-KXZGU	74	5442	Month-to-Month	DSL	3	62.45	Critical
0266-GMEAO	70	5879	Two Year	Fiber Optic	7	114.30	Critical
0270-THENM	77	6233	Two Year	DSL	4	69.85	Critical

5.1 Corrective Action Plan (OPL)

Priority	Action	Target Segment	Expected Impact	KPI to Measure	Timeline
CRITICAL	Bundle Online Security + Tech Support for all Fiber Optic customers	Fiber Optic · No Add-ons (372 customers)	Churn: 58.33% → 32.38% (-25.95 pts)	Fiber churn rate Add-on adoption rate	30 days
HIGH	Automated contract upgrade campaign triggered at Month 2 of tenure	Month-to-Month · 0–6 months + Replace Offer E with Offer A	M2M churn: 57.06% → ~0% (Two Year benchmark)	Contract conversion rate Early tenure retention	60 days
MEDIUM	Geo-targeted retention outreach + support quality audit in Suburban/Town	Suburban (30.72%) + Town (27.53%) + Senior 65+segment	Competitor-driven churn: 46.38% → target 35%	Satisfaction score by area Competitor loss rate	90 days

5.2 Financial Impact Simulation

The Financial Risk Optimization Playbook (Power BI Page 4) quantifies the projected savings from implementing all three corrective actions:

- **REC 1** — Bundle Savings: Fiber churn reduction from 58.33% to 32.38% across 372 at-risk customers
- **REC 2** — Contract Savings: Converting 20% of M2M early-tenure customers to annual contracts — eliminating the 57% dropout cliff
- **REC 3** — Offer Savings: Replacing Offer E (52.92% churn) with Offer A framework (6.73% churn)

5.3 Priority Customer Watchlist (OPL)

The OPL-style priority table in the dashboard identifies customers currently active but at critical churn risk — ranked by churn score and CLTV. Retention team should prioritize outreach to Critical and Hidden Risk segments within the next 30 days.

- Critical: High churn score + High CLTV → Immediate intervention required
- Hidden Risk: Low churn score + High CLTV → Most dangerous — metrics do not flag them, but financial exposure is highest
- High Risk: High churn score + Low CLTV → Monitor and apply low-cost retention offer
- Stable: Low churn score + Low CLTV → Maintain current service level

6. CONCLUSION

This quality metric analysis demonstrates that the 26.54% customer failure rate masks a more severe financial reality — a 30.50% revenue failure rate driven by the disproportionate exit of high-value customers. The data reveals three compounding risk factors operating simultaneously:

- A **demographic vulnerability**: Senior and Single segments with no natural lock-in, concentrated in early tenure stages
- A **service ecosystem gap**: Core services sold without protective add-ons, leaving customers unsupported during service disruptions
- A **contract structure misalignment**: Over 51% of customers on Month-to-Month contracts — the highest-risk retention vehicle in the portfolio

The corrective action plan targets these three root causes with measurable, time-bound interventions. The most critical insight from the survival curve analysis is that Two Year contract customers show 0.00% churn in the first 6 months — proving that the retention solution already exists within the current product portfolio. The priority is execution, not innovation.